

Vehicle Service Management System (VSM)

1. Authentication & Authorization

Functional Requirements

- FR-AUTH-01:** The system shall allow users to register with email and password (Customer self-registration).
- FR-AUTH-02:** The system shall allow staff (ServiceManager, Technician) to register with admin approval required.
- FR-AUTH-03:** The system shall allow users to log in using valid credentials.
- FR-AUTH-04:** The system shall generate a JWT token upon successful login.
- FR-AUTH-05:** The system shall restrict API access based on user roles.
- FR-AUTH-06:** The system shall allow users to log out by invalidating the session on client side.
- FR-AUTH-07:** The system shall send OTP for email verification upon registration.
- FR-AUTH-08:** The system shall verify email using 6-digit OTP within 10-minute expiry.
- FR-AUTH-09:** The system shall allow users to resend OTP for email verification.
- FR-AUTH-10:** The system shall allow users to change their password (authenticated).
- FR-AUTH-11:** The system shall allow users to request password reset link (forgot password).
- FR-AUTH-12:** The system shall allow password reset using token from email link.

Business Rules

- Password must be stored in hashed format (BCrypt)
- JWT token must include user role, user ID, and email
- Unauthorized users must not access protected APIs
- OTP is valid for 10 minutes only
- Maximum 5 OTP verification attempts allowed
- Password reset token expires in 30 minutes
- Staff registration requires admin approval before login is allowed
- Email must be verified before user can log in

2. User & Role Management (Admin)

Functional Requirements

FR-ADMIN-01: Admin shall be able to create staff users (ServiceManager, Technician).

FR-ADMIN-02: Admin shall be able to assign roles (ServiceManager / Technician).

FR-ADMIN-03: Admin shall be able to view all users.

FR-ADMIN-04: Admin shall be able to activate or deactivate users.

FR-ADMIN-05: Admin shall be able to update user information.

FR-ADMIN-06: Admin shall be able to delete staff users (ServiceManager, Technician only).

FR-ADMIN-07: Admin shall be able to view all technicians with workload information.

FR-ADMIN-08: Admin shall be able to view pending staff registration approvals.

FR-ADMIN-09: Admin shall be able to approve pending staff registrations.

FR-ADMIN-10: Admin shall be able to reject pending staff registrations.

FR-ADMIN-11: ServiceManager shall be able to view user by ID.

Business Rules

- Only Admin can manage users
- Each user must have exactly one role
- Admin and Customer accounts cannot be deleted
- Only ServiceManager and Technician accounts can be created by Admin
- Customers must register themselves
- Only one Admin account exists in the system

User Roles

Role	Description
Admin	System administrator with full access
ServiceManager	Manages service requests, assignments, billing
Technician	Performs vehicle servicing tasks
Customer	Vehicle owner who requests services

3. Vehicle Management (Customer)

Functional Requirements

FR-VEH-01: Customer shall be able to add new vehicles to their account.

FR-VEH-02: Customer shall be able to view their own vehicles.

FR-VEH-03: Customer shall be able to update their vehicle information.

FR-VEH-04: Customer shall be able to delete their vehicles (if no service requests exist).

FR-VEH-05: ServiceManager shall be able to view all vehicles.

Vehicle Attributes

Attribute	Description
VehicleId	Unique identifier
CustomerId	Owner's user ID
RegistrationNumber	Vehicle registration number
Make	Vehicle manufacturer (e.g., Toyota)
Model	Vehicle model (e.g., Camry)
Year	Manufacturing year
VehicleType	Type (e.g., Car, SUV, Motorcycle, Truck)
FuelType	Fuel type (Petrol, Diesel, Electric, Hybrid, CNG)
Color	Vehicle color
RcNumber	RC registration number
CreatedAt	Registration timestamp

Business Rules

- Vehicle registration number must be unique
- Cannot delete vehicle with existing service requests
- Customer can only manage their own vehicles

4. Service Category Management (Admin/ServiceManager)

Functional Requirements

FR-CAT-01: Admin/ServiceManager shall be able to create service categories.

FR-CAT-02: Admin/ServiceManager shall be able to define base price for each category.

FR-CAT-03: Admin/ServiceManager shall be able to define estimated duration in minutes.

FR-CAT-04: Admin/ServiceManager shall be able to activate or deactivate categories.

FR-CAT-05: Anyone shall be able to view all service categories.

FR-CAT-06: Anyone shall be able to view only active service categories.

FR-CAT-07: Admin/ServiceManager shall be able to update service categories.

FR-CAT-08: Admin/ServiceManager shall be able to delete service categories.

Service Category Attributes

Attribute	Description
CategoryId	Unique identifier
CategoryName	Name of service category
Description	Detailed description
BasePrice	Standard price for the service
EstimatedDurationMinutes	Expected time to complete
IsActive	Active status flag

Business Rules

- Base price must be a positive value
- Estimated duration must be greater than zero
- Only active categories can be selected for new service requests

5. Service Request Management

Functional Requirements

FR-SR-01: Customer shall be able to create a service request for their vehicle.

FR-SR-02: Customer shall be able to select one or more service categories while creating request.

FR-SR-03: Customer shall be able to provide issue description.

FR-SR-04: Customer shall be able to set priority (Normal/Urgent).

- FR-SR-05:** Customer shall be able to schedule service date.
- FR-SR-06:** Customer shall be able to add customer remarks.
- FR-SR-07:** System shall store service request with initial status "Requested".
- FR-SR-08:** Customer shall be able to view their own service requests.
- FR-SR-09:** Customer shall be able to cancel their service request (with reason).
- FR-SR-10:** Customer shall be able to reschedule their service request.
- FR-SR-11:** Admin/ServiceManager shall be able to view all service requests.
- FR-SR-12:** Admin/ServiceManager shall be able to update service request status.
- FR-SR-13:** Admin/ServiceManager shall be able to update service request details.
- FR-SR-14:** Technician shall be able to view their assigned service requests.
- FR-SR-15:** Technician shall be able to add technician remarks.
- FR-SR-16:** System shall parse issue description into individual tasks.
- FR-SR-17:** System shall support multiple service categories per request (ServiceRequestCategory junction).
- FR-SR-18:** ServiceManager shall be able to adjust price per category.

Service Request Attributes

Attribute	Description
ServiceRequestId	Unique identifier
VehicleId	Associated vehicle
CustomerId	Customer who made request
CategoryId	Primary service category (optional)
IssueDescription	Customer's description of issues
Priority	Normal / Urgent
Status	Current status in lifecycle
RequestedDate	When request was made
ScheduledDate	Scheduled service date
CompletedDate	When service was completed
CustomerRemarks	Additional notes from customer

TechnicianRemarks	Notes from technician
EstimatedCost	Estimated service cost
ActualCost	Final service cost

Business Rules

- Customer can only create requests for their own vehicles
- Customer can only create requests for active service categories
- Customer can only view/cancel/reschedule their own requests
- Cannot cancel completed or closed requests
- Reschedule date must be in the future
- Issue description is automatically parsed into service tasks

6. Service Assignment & Technician Workflow

Functional Requirements

FR-ASGN-01: ServiceManager shall be able to assign technician to service request.

FR-ASGN-02: ServiceManager shall be able to view all assignments.

FR-ASGN-03: ServiceManager shall be able to update assignments (reassign technician).

FR-ASGN-04: ServiceManager shall be able to delete assignments (not completed).

FR-ASGN-05: Technician shall be able to view their own assignments.

FR-ASGN-06: Technician shall be able to start work (update status to InProgress).

FR-ASGN-07: Technician shall be able to complete work (update status to Completed).

FR-ASGN-08: Technician shall be able to reopen completed work (if not closed by manager).

FR-ASGN-09: Technician shall be able to add notes to assignment.

FR-ASGN-10: System shall record starting date when work begins.

FR-ASGN-11: System shall record completed date when work finishes.

FR-ASGN-12: System shall automatically update service request status on assignment status change.

Service Assignment Attributes

Attribute	Description
AssignmentId	Unique identifier
ServiceRequestId	Associated service request
TechnicianId	Assigned technician
AssignedDate	When assignment was created
StartedDate	When technician started work
CompletedDate	When work was completed
Status	Assigned / InProgress / Completed
Notes	Technician's notes

Assignment Status Flow

Assigned → InProgress → Completed

Business Rules

- Only ServiceManager can create assignments
- Service request must exist and be in "Requested" status for assignment
- Technicians must exist and have Technician role
- Service request can only have one active assignment
- Cannot delete completed assignments
- Only assigned technicians can update their assignment status
- Cannot reopen if service request is already closed

7. Service Tasks Management

Functional Requirements

FR-TASK-01: System shall automatically create tasks from issue description (comma-separated items).

FR-TASK-02: Technician shall be able to view all tasks for a service request.

FR-TASK-03: Technician shall be able to toggle task completion status.

FR-TASK-04: Admin/ServiceManager shall be able to update task prices.

FR-TASK-05: System shall track who completed each task.

FR-TASK-06: System shall track when each task is completed.

FR-TASK-07: System shall maintain task order.

Service Task Attributes

Attribute	Description
ServiceTaskId	Unique identifier
ServiceRequestId	Associated service request
Description	Task description
IsCompleted	Completion status
CompletedDate	When task was completed
CompletedByUserId	Technician who completed
Price	Task price for billing
OrderIndex	Display order
CreatedDate	When task was created

Business Rules

- Only assigned technician can toggle task completion
- Task price can only be updated by Admin/ServiceManager
- Completing all tasks updates service request ActualCost

8. Parts & Inventory Management

Functional Requirements

FR-PART-01: Admin/ServiceManager shall be able to add new parts to inventory.

FR-PART-02: Admin/ServiceManager shall be able to update part information.

FR-PART-03: Admin/ServiceManager shall be able to delete parts.

FR-PART-04: Admin/ServiceManager/Technician shall be able to view all parts.

FR-PART-05: Admin/ServiceManager/Technician shall be able to view part by ID.

FR-PART-06: Admin/ServiceManager shall be able to view parts below reorder level.

FR-PART-07: System shall track quantity in stock.

FR-PART-08: System shall track reorder level for low stock alerts.

Part Attributes

Attribute	Description
PartId	Unique identifier
PartName	Name of the part
PartNumber	Manufacturer part number
Description	Part description
UnitPrice	Price per unit
QuantityInStock	Current stock quantity
ReorderLevel	Minimum stock threshold
Supplier	Supplier name

Business Rules

- Part number must be unique
- Quantity in stock cannot be negative
- Low stock alert triggered when quantity $\leq$ reorder level
- Cannot delete part if used in active service requests

9. Part Orders (Supplier Orders)

Functional Requirements

FR-PORD-01: ServiceManager shall be able to create part orders from suppliers.

FR-PORD-02: ServiceManager shall be able to view all part orders.

FR-PORD-03: ServiceManager shall be able to view order by ID.

FR-PORD-04: ServiceManager shall be able to view pending orders.

FR-PORD-05: ServiceManager shall be able to view orders by part.

FR-PORD-06: ServiceManager shall be able to update order details.

FR-PORD-07: ServiceManager shall be able to mark order as delivered.

FR-PORD-08: ServiceManager shall be able to cancel orders.

FR-PORD-09: System shall auto-generate order number.

FR-PORD-10: System shall calculate total amount (quantity $\times$ unit price).

FR-PORD-11: System shall update part stock when order is marked delivered.

Part Order Attributes

Attribute	Description
PartOrderId	Unique identifier
OrderNumber	Auto-generated order number
PartId	Part being ordered
Quantity	Order quantity
UnitPrice	Price per unit
TotalAmount	Total order amount
Supplier	Supplier name
Status	Pending / Ordered / Shipped / Delivered / Cancelled
OrderDate	When order was placed
ExpectedDeliveryDate	Expected delivery
DeliveredDate	Actual delivery date
OrderedByUserId	User who placed order
Notes	Order notes

Business Rules

- Only Admin/ServiceManager can manage part orders
- Cannot cancel delivered orders
- Stock is automatically updated when order is marked delivered
- Order numbers are system-generated and unique

10. Service Parts (Parts Used in Service)

Functional Requirements

FR-SP-01: ServiceManager/Technician shall be able to add parts to service request.

FR-SP-02: ServiceManager/Technician shall be able to remove parts from service request.

FR-SP-03: ServiceManager/Technician shall be able to view parts used in service request.

FR-SP-04: System shall calculate total parts cost for service request.

FR-SP-05: System shall auto-deduct parts from inventory when added to service.

FR-SP-06: System shall return parts to inventory when removed from service.

Service Part Attributes

Attribute	Description
ServicePartId	Unique identifier
ServiceRequestId	Associated service request
PartId	Part used
QuantityUsed	Quantity used
UnitPrice	Price at time of use
TotalPrice	Quantity × Unit Price

Business Rules

- Cannot add part if insufficient stock
- Part price is captured at time of adding to service
- Removing part returns quantity to inventory

11. Billing & Invoicing

Functional Requirements

FR-BILL-01: ServiceManager shall be able to generate bill for completed service.

FR-BILL-02: System shall auto-calculate service charge from tasks and categories.

FR-BILL-03: System shall auto-calculate parts charge from service parts.

FR-BILL-04: System shall apply tax percentage.

FR-BILL-05: System shall apply discount.

FR-BILL-06: System shall generate unique bill number.

FR-BILL-07: ServiceManager shall be able to view all bills.

FR-BILL-08: ServiceManager shall be able to view bill by ID.

FR-BILL-09: Customer shall be able to view their own bills.

FR-BILL-10: ServiceManager shall be able to update bill.

FR-BILL-11: ServiceManager shall be able to recalculate bill totals.

FR-BILL-12: Admin shall be able to delete bills (no payments).

FR-BILL-13: System shall filter bills by status, customer, date range, overdue status.

Bill Attributes

Attribute	Description
BillId	Unique identifier
ServiceRequestId	Associated service request
ServiceCharge	Total service/labor charge
PartsCharge	Total parts charge
Tax	Tax amount
Discount	Discount amount
TotalAmount	Final amount (Service + Parts + Tax - Discount)
GeneratedDate	Bill generation date
BillNumber	Unique bill number

Bill Calculation Formula

$$\text{TotalAmount} = (\text{ServiceCharge} + \text{PartsCharge} + \text{Tax}) - \text{Discount}$$

Business Rules

- One bill per service request
- Cannot generate bill if one already exists
- Cannot delete bill with existing payments

12. Payment Processing & Tracking

Functional Requirements

FR-PAY-01: ServiceManager shall be able to record payments.

FR-PAY-02: Customer shall be able to make payments for their own bills.

FR-PAY-03: System shall track payment date.

FR-PAY-04: System shall update payment status.

FR-PAY-05: System shall support multiple payment methods.

FR-PAY-06: System shall store transaction ID.

FR-PAY-07: System shall track outstanding balance.

FR-PAY-08: System shall support partial payments.

FR-PAY-09: User shall be able to view payments by bill.

Payment Attributes

Attribute	Description
PaymentId	Unique identifier
BillId	Associated bill
AmountPaid	Payment amount
PaymentDate	When payment was made
PaymentMethod	Cash / Card / UPI / NetBanking
TransactionId	Transaction reference
PaymentStatus	Pending / Paid

Payment Status Values

Status	Description
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Pending	Payment initiated but not confirmed
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Paid	Full payment received
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Business Rules

- Customer can only pay for their own bills
- Payment amount cannot exceed outstanding balance
- Multiple payments allowed until bill is fully paid
- System updates bill status based on payments

13. In-App Notifications System

Functional Requirements

FR-NOTIF-01: System shall send in-app notifications for key events.

FR-NOTIF-02: User shall be able to view their notifications.

FR-NOTIF-03: User shall be able to get unread notification count.

FR-NOTIF-04: User shall be able to mark notification as read.

FR-NOTIF-05: User shall be able to mark all notifications as read.

FR-NOTIF-06: System shall support user-specific notifications.

FR-NOTIF-07: System shall support role-based notifications.

FR-NOTIF-08: System shall paginate notification retrieval.

FR-NOTIF-09: System shall support filtering for unread only.

In-App Notification Attributes

Attribute	Description
Id	Unique identifier (GUID)
Title	Short notification title
Message	Detailed message
Type	Notification type
TargetUserId	Specific user (optional)
TargetRole	Target role (optional)
IsRead	Read status
CreatedAt	Creation timestamp
CreatedByUserId	Who triggered notification
RelatedEntityId	Related entity for navigation
RelatedEntityType	Entity type (ServiceRequest, Bill, etc.)
ActionUrl	URL for click action

Notification Types

Type	Description
ServiceRequestCreated	New service request created
ServiceRequestUpdated	Service request updated

ServiceRequestCancelled	Service request cancelled
ServiceRequestCompleted	Service completed
ServiceRequestRescheduled	Service rescheduled
ServiceRequestClosed	Service request closed
TechnicianAssigned	Technician assigned to service
AssignmentStarted	Technician started work
AssignmentCompleted	Technician completed work
ServiceReopened	Service reopened
BillGenerated	Bill generated
PaymentReceived	Payment received
PaymentOverdue	Payment overdue
PaymentRequired	Payment required notification
PartOrderCreated	Part order created
PartOrderReceived	Part order received
LowStockAlert	Part stock below reorder level
AccountApproved	Staff account approved
AccountRejected	Staff account rejected
PasswordChanged	Password changed
StaffApprovalRequested	New staff pending approval

Business Rules

- Notifications can target specific user OR role (not both)
- Read status is tracked per notification
- Notifications ordered by creation date (newest first)

14. Email Notifications System

Functional Requirements

FR-EMAIL-01: System shall send email for service booking confirmation.

FR-EMAIL-02: System shall send email for service status updates.

FR-EMAIL-03: System shall send email for staff approval notifications.

FR-EMAIL-04: System shall send email for technician assignment.

FR-EMAIL-05: System shall send email for service completion.

FR-EMAIL-06: System shall send email for billing notifications.

FR-EMAIL-07: System shall send email for OTP verification.

FR-EMAIL-08: System shall send email for password reset.

FR-EMAIL-09: System shall process emails via background service queue.

Email Notification Types

Type	Description
ServiceBookingConfirmation	Confirm service request creation
ServiceStatusUpdate	Service status changed
StaffApproval	Staff registration status
TechnicianAssignment	Technician assigned notification
ServiceCompletion	Service completed
Billing	Bill and payment notifications
EmailVerification	OTP for email verification
PasswordReset	Password reset link

Business Rules

- Emails processed asynchronously via queue
- SMTP configuration required
- Failed emails logged but don't block operations

15. Service Request Status & Lifecycle Tracking

Functional Requirements

FR-STATUS-01: System shall track service request status.

FR-STATUS-02: Customer shall be able to view service status.

FR-STATUS-03: System shall update status automatically based on actions.

FR-STATUS-04: System shall maintain status history.

Status Definitions

Status	Description
Requested	Customer has submitted request
Assigned	Technician has been assigned
InProgress	Technician has started work
Completed	Technician has completed work
Closed	ServiceManager has closed/finalized
Cancelled	Customer has cancelled request

Business Rules

- Status can only progress forward (except reopening)
- Cancelled status is final
- Closed status is final
- Technician completing work sets Completed status
- ServiceManager closes completed requests

16. Security Requirements (Non-Functional)

Security Requirements

NFR-SEC-01: APIs must be secured using JWT.

NFR-SEC-02: Role-based authorization must be enforced.

NFR-SEC-03: Sensitive data must not be exposed (passwords, OTP hashes).

NFR-SEC-04: Unauthorized access must be denied.

NFR-SEC-05: Passwords must be hashed using BCrypt.

NFR-SEC-06: OTPs must be stored hashed.

NFR-SEC-07: Password reset tokens must be stored hashed.

NFR-SEC-08: API responses must use consistent format.

17. Performance & Quality Requirements

Performance Requirements

NFR-PERF-01: System should handle multiple concurrent users.

NFR-PERF-02: API responses should be optimized using LINQ.

NFR-PERF-03: Database queries should be efficient.

NFR-PERF-04: Email notifications processed asynchronously.

NFR-PERF-05: In-app notifications processed asynchronously.

Quality Requirements

NFR-QUAL-01: Code should be modular and maintainable.

NFR-QUAL-02: Proper validations and error handling required.

NFR-QUAL-03: Repository pattern must be used for data access.

NFR-QUAL-04: Unit of Work pattern for transactions.

NFR-QUAL-05: DTOs for data transfer (no entity exposure).

NFR-QUAL-06: AutoMapper for entity-DTO mapping.

NFR-QUAL-07: Global exception handling middleware.

NFR-QUAL-08: Consistent API response format.

NFR-QUAL-09: API documentation via Swagger/OpenAPI.

18. Service Change History & Tracking

Functional Requirements

FR-HIST-01: System shall track service request changes.

FR-HIST-02: System shall record action type (Created, Updated, Assigned, Completed).

FR-HIST-03: System shall record field name, old value, and new value.

FR-HIST-04: System shall record timestamp of change.

FR-HIST-05: System shall store changes via background service queue.

Service Change History Attributes

Attribute	Description
Id	Unique identifier

ServiceRequestId	Associated service request
Action	Created / Updated / Assigned / Completed
FieldName	Changed field name
OldValue	Previous value
NewValue	New value
ChangedOn	Change timestamp

Business Rules

- History is append-only (no updates/deletes)
- Changes processed asynchronously via queue
- In-memory event store for recent events (last 100)